

A Surgery Formula for Seiberg–Witten Invariants

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Abstract

This report explains the surgery formula for Seiberg–Witten invariants of smooth 4-manifolds with first Betti number one. Let X be a closed oriented smooth 4-manifold with $H^1(X; \mathbb{Z}) \cong \mathbb{Z}$, let an embedded framed loop γ pair to one with its generator, and let X' be obtained by replacing $S^1 \times D^3$ with $D^2 \times S^2$. For a Spin^c structure $\mathfrak{s}$ with one-dimensional moduli space, the circle direction defines a degree-one class Θ on the irreducible configuration quotient. Qiu’s formula is

$$\text{SW}_\Theta(X, \mathfrak{s}) = \text{SW}(X', \mathfrak{s}'),$$

where $\mathfrak{s}'$ is the unique corresponding structure on X' . I explain the surgery topology, the cylindrical-end gluing input, and the orientation of the resulting count, followed by applications to exotic smooth structures and adjunction inequalities. Finally, I compare this formula with the earlier reverse sphere-surgery identity of Ozsváth and Szabó, which allows more general insertions and several primitive loops, and discuss its stronger surface-insertion bound.

Contents

1	Introduction	2
2	Review of the Seiberg–Witten setup	3
2.1	Spin-c structures and the Seiberg–Witten equations	3
2.2	The topology of the configuration quotient when $b_1=1$	4
2.3	Cut-down invariants	5
3	Topology of the 1-surgery	6
3.1	The decomposition	6
3.2	Extension of the Spin-c structure	6
3.3	Change of expected dimension	7
4	Cylindrical ends and the gluing theorem	8
4.1	Cylindrical-end notation	8
4.2	The gluing theorem used here	9
4.3	Verification for the surgery pieces	10
5	The two global gluing identifications	11
5.1	Gluing back $S^1 \times D^3$	11
5.2	Gluing in $D^2 \times S^2$	12

6	Proof of the surgery formula	12
6.1	The boundary circle is the Theta direction	12
6.2	Counting the points after surgery	13
6.3	Orientation and sign	13
6.4	Higher odd-dimensional moduli spaces	14
7	Applications	14
7.1	Exotic smooth structures after adding $S^1 \times S^3$	14
7.2	Elliptic surfaces and knot surgery	15
7.3	Odd-dimensional basic classes	16
7.4	Adjunction inequality	16
8	The earlier surgery identity and its wider scope	17
8.1	Why a primitive loop gives a trivial core sphere	18
8.2	The formula with insertions	18
8.3	Several loops and the applications	19

1 Introduction

Let X be a closed oriented smooth 4-manifold and let $\gamma \subset X$ be an embedded loop with a chosen framing of its normal bundle. A 1-surgery along γ means that we remove a tubular neighborhood

$$N(\gamma) \cong S^1 \times D^3$$

and glue back $D^2 \times S^2$ along the common boundary $S^1 \times S^2$. If

$$X_0 = X \setminus \text{int}(S^1 \times D^3), \quad Y = \partial X_0 = S^1 \times S^2,$$

then

$$X = X_0 \cup_Y (S^1 \times D^3), \quad X' = X_0 \cup_Y (D^2 \times S^2).$$

The topological operation kills the free homology direction represented by γ . The framing fixes the boundary identification; we use the product framing throughout the gluing argument. The point of Qiu's result is that there is a completely parallel operation on the Seiberg–Witten moduli space [1, Theorem 3.17].

Assume throughout that

$$H^1(X; \mathbb{Z}) \cong \mathbb{Z}$$

and that γ pairs to 1 with a fixed generator of $H^1(X; \mathbb{Z})$. Let $\mathfrak{s}$ be a Spin^c structure on X , with determinant line bundle L . Its expected Seiberg–Witten dimension is

$$d(X, \mathfrak{s}) = \frac{1}{4} \left(c_1(L)^2 - 2\chi(X) - 3\sigma(X) \right). \quad (1)$$

The main case is

$$d(X, \mathfrak{s}) = 1.$$

After choosing a generic metric and perturbation, the moduli space $\mathcal{M}(X, \mathfrak{s})$ is then a compact oriented 1-manifold. The orientation requires a homology orientation of $H^1(X; \mathbb{R}) \oplus H_+^2(X; \mathbb{R})$; the same choice is carried through the surgery.

When $b_1(X) = 1$, the configuration quotient has one degree-one cohomology class coming from the $H^1(X)$ direction. Denote its generator by

$$\Theta \in H^1(\mathcal{B}^*(X, \mathfrak{s}); \mathbb{Z}).$$

The corresponding cut-down invariant is

$$\text{SW}_\Theta(X, \mathfrak{s}) = \langle \Theta, [\mathcal{M}(X, \mathfrak{s})] \rangle.$$

The surgery formula says that this odd-dimensional invariant is an ordinary zero-dimensional Seiberg–Witten invariant after surgery. In the notation used here, this is Qiu’s main surgery theorem [1, Theorems 1.1 and 3.17].

Theorem 1.1 (Qiu’s surgery formula). *Let X be a closed connected oriented smooth 4-manifold with $b_2^+(X) > 1$ and $H^1(X; \mathbb{Z}) \cong \mathbb{Z}$. Let $\gamma \subset X$ be an embedded framed loop whose homology class is primitive in $H_1(X; \mathbb{Z})/\text{tors}$, and let X' be obtained by 1-surgery along γ . Suppose $\mathfrak{s}$ is a Spin^c structure with*

$$d(X, \mathfrak{s}) = 1.$$

Then $\mathfrak{s}|_{X_0}$ extends uniquely up to isomorphism to a Spin^c structure $\mathfrak{s}'$ on X' , and, with compatible homology orientations,

$$\boxed{\text{SW}_\Theta(X, \mathfrak{s}) = \text{SW}(X', \mathfrak{s}').}$$

The geometric reason, as explained in Qiu’s introduction, is already visible from the three simple pieces [1, pp. 2–3]. With the metrics used in the proof,

$$\mathcal{M}(S^1 \times S^2) \cong S^1, \quad \mathcal{M}(S^1 \times D^3) \cong S^1, \quad \mathcal{M}(D^2 \times S^2) \cong \{\text{pt}\}.$$

The asymptotic-value map from $S^1 \times D^3$ to $S^1 \times S^2$ identifies their circles of reducibles, whereas the map from $D^2 \times S^2$ selects one point of the boundary circle. Thus replacing $S^1 \times D^3$ by $D^2 \times S^2$ replaces one circle-valued boundary condition by a point condition. The class Θ measures exactly this same circle direction. The rest of the proof is devoted to making this picture rigorous.

2 Review of the Seiberg–Witten setup

I first recall the minimum amount of Seiberg–Witten theory needed later. The purpose is not to reconstruct the theory from the beginning, but to fix the notation and explain where the degree-one class Θ comes from. The conventions agree with Qiu’s setup [1, Sections 1.1 and 2.2].

2.1 Spin^c structures and the Seiberg–Witten equations

A Spin^c structure $\mathfrak{s}$ on an oriented Riemannian 4-manifold X gives rank-two complex spinor bundles

$$S_\mathfrak{s}^+, \quad S_\mathfrak{s}^-$$

and a determinant line bundle

$$L = \det S_\mathfrak{s}^+ = \det S_\mathfrak{s}^-.$$

The first Chern class $c_1(L)$ is characteristic:

$$c_1(L) \equiv w_2(TX) \pmod{2}.$$

Conversely, the set of Spin^c structures is a torsor over $H^2(X; \mathbb{Z})$. If $E \rightarrow X$ is a complex line bundle, then

$$\mathfrak{s} \otimes E$$

is another Spin^c structure and

$$\det(\mathfrak{s} \otimes E) = \det(\mathfrak{s}) \otimes E^2.$$

In particular,

$$c_1(\det(\mathfrak{s} \otimes E)) = c_1(L) + 2c_1(E).$$

This torsor description will be the key topological input when we extend Spin^c structures across the surgery.

Fix a unitary connection A on L and a positive spinor $\Phi \in \Gamma(S_s^+)$. The Seiberg–Witten equations, with a real self-dual perturbation $\eta \in \Omega_+^2(X; \mathbb{R})$, have the form

$$\begin{cases} D_A \Phi = 0, \\ F_A^+ + i\eta = q(\Phi), \end{cases} \quad (2)$$

where $D_A : \Gamma(S_s^+) \rightarrow \Gamma(S_s^-)$ is the Dirac operator determined by A and the Levi-Civita connection, while $q : \Gamma(S_s^+) \rightarrow i\Omega_+^2(X)$ is the standard quadratic map. With this convention both F_A^+ and $i\eta$ are imaginary-valued. Changing the sign of the perturbation would amount to replacing η by $-\eta$ throughout.

The gauge group is

$$\mathcal{G}(X) = \text{Map}(X, S^1).$$

With the convention that $u \in \mathcal{G}(X)$ acts on spinors by $u\Phi$, its action on a determinant connection is $A \mapsto A - 2u^{-1}du$. The factor 2 reflects the determinant representation of $\text{Spin}^c(4)$. Write

$$\mathbb{C}(X, \mathfrak{s}) = \mathcal{A}(L) \times \Gamma(S_s^+), \quad \mathcal{B}^*(X, \mathfrak{s}) = \mathbb{C}^*(X, \mathfrak{s}) / \mathcal{G}(X),$$

where the star means that the spinor is not identically zero. The gauge action on $\mathbb{C}^*$ is free; the full quotient $\mathcal{B} = \mathbb{C} / \mathcal{G}$ also contains reducible configurations and does not have the same homotopy type. The moduli space $\mathcal{M}(X, \mathfrak{s})$ is the set of gauge-equivalence classes of solutions of (2). A solution is *reducible* when $\Phi \equiv 0$ and *irreducible* otherwise, so the generic moduli spaces below lie in $\mathcal{B}^*$.

For a generic metric and perturbation, reducibles are absent when $b_2^+(X) > 0$, and the irreducible moduli space is smooth of the expected dimension (1). The stronger assumption $b_2^+(X) > 1$ makes its signed intersection numbers independent of the metric and perturbation. Compactness is the usual Seiberg–Witten compactness, so in the one-dimensional case the moduli space is a finite union of oriented circles.

2.2 The topology of the configuration quotient when $b_1 = 1$

The component group of the gauge group is

$$\pi_0 \mathcal{G}(X) \cong H^1(X; \mathbb{Z}).$$

The space $\mathbb{C}^*$ of irreducible configurations is contractible up to homotopy: its connection factor is affine, and the complement of the zero section in the infinite-dimensional spinor space is contractible. Since the gauge action there is free, $\mathcal{B}^*$ has the homotopy type of $B\mathcal{G}(X)$. Its homotopy groups record $\pi_0 \mathcal{G}(X) = H^1(X; \mathbb{Z})$ and $\pi_1 \mathcal{G}(X) = H^0(X; \mathbb{Z}) \cong \mathbb{Z}$. In particular, when

$$H^1(X; \mathbb{Z}) \cong \mathbb{Z},$$

its homotopy type is

$$\mathcal{B}^*(X, \mathfrak{s}) \simeq \mathbb{CP}^\infty \times S^1.$$

This describes the space of irreducibles; using the full configuration quotient here would lose the $\mathbb{CP}^\infty$ factor. Thus there are two natural universal cohomology classes, as used in the definition of the cut-down invariant in [1, Section 1.1]:

$$U \in H^2(\mathcal{B}^*; \mathbb{Z}), \quad \Theta \in H^1(\mathcal{B}^*; \mathbb{Z}),$$

where U is the generator from $\mathbb{CP}^\infty$ and Θ is the generator from S^1 .

There is also a more geometric description of the degree-one class. Choose the loop γ pairing to 1 with the fixed generator of $H^1(X; \mathbb{Z})$. Fix a reference determinant connection A_0 . The half-difference $(A - A_0)/2$ is a globally defined imaginary one-form. Its exponential integral along γ is gauge invariant: a gauge transformation changes that integral by an integral multiple of $2\pi i$. This defines a normalized circle coordinate h_γ ; changing A_0 only rotates the target. Its cohomology class is the usual degree-one insertion $\mu(\gamma)$ [2, Proposition 9.1]. If one used the raw holonomy of A instead, the induced class could be twice the primitive class because A changes by $-2u^{-1}du$. Thus

$$h_\gamma : \mathcal{B}^*(X, \mathfrak{s}) \longrightarrow S^1,$$

and if η_{S^1} denotes the positive generator of $H^1(S^1; \mathbb{Z})$, then

$$\Theta = h_\gamma^* \eta_{S^1}.$$

This is the interpretation we will use in the final proof: Θ detects the normalized holonomy around the circle killed by surgery. Fixing the sign of α and orienting the holonomy circle accordingly fixes the sign of Θ .

2.3 Cut-down invariants

If $d(X, \mathfrak{s}) = 0$, the usual invariant is the signed count

$$\text{SW}(X, \mathfrak{s}) = \#\mathcal{M}(X, \mathfrak{s}).$$

If $d(X, \mathfrak{s}) = 1$, then there is no signed zero-dimensional count before inserting a class. Instead one pairs the fundamental class of the moduli space with Θ :

$$\text{SW}_\Theta(X, \mathfrak{s}) = \langle \Theta, [\mathcal{M}(X, \mathfrak{s})] \rangle.$$

Equivalently, if

$$\rho : \mathcal{M}(X, \mathfrak{s}) \longrightarrow S^1$$

represents Θ and $p \in S^1$ is a regular value, then

$$\text{SW}_\Theta(X, \mathfrak{s}) = \deg \rho = \#\rho^{-1}(p),$$

where the last count uses the orientation induced from $\mathcal{M}(X, \mathfrak{s})$ and S^1 . The whole surgery formula will amount to proving that the zero-dimensional moduli space on X' is exactly this inverse image.

3 Topology of the 1-surgery

3.1 The decomposition

Fix a generator

$$\alpha \in H^1(X; \mathbb{Z})$$

and choose $\gamma \subset X$ so that

$$\langle \alpha, [\gamma] \rangle = 1.$$

Let

$$N = S^1 \times D^3$$

be a sufficiently small framed tubular neighborhood of γ . Set

$$X_0 = X \setminus \text{int } N, \quad Y = \partial X_0 = S^1 \times S^2.$$

Then

$$X = X_0 \cup_Y N, \quad X' = X_0 \cup_Y (D^2 \times S^2).$$

The primitive assumption is essential: it says that an integral class pairs to 1 with $[\gamma]$. It makes $H^1(X_0; \mathbb{Z}) \rightarrow H^1(Y; \mathbb{Z})$ onto and, in the relevant exact sequences, identifies $H^2(X_0; \mathbb{Z})$ with H^2 of both closed manifolds. If the loop is homologically trivial, the boundary circle need not extend to an integral cohomology class on X_0 ; different extensions of a Spin^c structure can then occur.

3.2 Extension of the Spin^c structure

Let $\mathfrak{s}$ be a Spin^c structure on X and put

$$\mathfrak{s}_0 = \mathfrak{s}|_{X_0}.$$

We now prove carefully that $\mathfrak{s}_0$ extends uniquely over X' . This is the topological statement proved in Qiu's Theorem 2.9 [1].

Proposition 3.1. *There is a unique Spin^c structure $\mathfrak{s}'$ on X' satisfying*

$$\mathfrak{s}'|_{X_0} \cong \mathfrak{s}_0.$$

If $L = \det \mathfrak{s}$ and $L' = \det \mathfrak{s}'$, then

$$\langle c_1(L')^2, [X'] \rangle = \langle c_1(L)^2, [X] \rangle. \quad (3)$$

Proof. The set of isomorphism classes of Spin^c structures on a fixed oriented manifold is a free and transitive $H^2(-; \mathbb{Z})$ -set, even when this group has torsion. Thus if $\mathfrak{t}$ is any Spin^c structure on X' , then over X_0 there is a complex line bundle E_0 such that

$$\mathfrak{t}|_{X_0} \otimes E_0 \cong \mathfrak{s}_0.$$

It is therefore enough to show that restriction $H^2(X'; \mathbb{Z}) \rightarrow H^2(X_0; \mathbb{Z})$ is an isomorphism. This also resolves possible torsion ambiguities that would be invisible from determinant Chern classes alone.

Consider the Mayer–Vietoris sequence for

$$X' = X_0 \cup_Y (D^2 \times S^2), \quad Y = S^1 \times S^2.$$

The relevant part is

$$\begin{aligned} H^1(X_0) \oplus H^1(D^2 \times S^2) &\longrightarrow H^1(Y) \longrightarrow H^2(X') \\ &\longrightarrow H^2(X_0) \oplus H^2(D^2 \times S^2) \longrightarrow H^2(Y). \end{aligned}$$

Here

$$H^1(D^2 \times S^2) = 0, \quad H^1(Y) \cong \mathbb{Z},$$

and the restriction

$$H^1(X_0) \longrightarrow H^1(Y)$$

is surjective because the boundary S^1 is the same homology direction as γ . Thus the connecting map $H^1(Y) \rightarrow H^2(X')$ is zero.

On the right, the restriction

$$H^2(D^2 \times S^2) \longrightarrow H^2(S^1 \times S^2)$$

is an isomorphism. Consequently the kernel of

$$H^2(X_0) \oplus H^2(D^2 \times S^2) \longrightarrow H^2(Y)$$

is the graph of a unique homomorphism from $H^2(X_0)$ to $H^2(D^2 \times S^2)$. Exactness therefore gives an isomorphism

$$H^2(X'; \mathbb{Z}) \xrightarrow{\cong} H^2(X_0; \mathbb{Z}). \quad (4)$$

Hence $c_1(E_0)$ extends uniquely to a class in $H^2(X'; \mathbb{Z})$, so E_0 extends uniquely up to isomorphism to a line bundle E' on X' . Then

$$\mathfrak{s}' = \mathfrak{t} \otimes E'$$

is an extension of $\mathfrak{s}_0$.

For uniqueness, suppose $\mathfrak{s}'_1$ and $\mathfrak{s}'_2$ both restrict to $\mathfrak{s}_0$. Then

$$\mathfrak{s}'_2 \cong \mathfrak{s}'_1 \otimes E$$

for a line bundle E on X' . Restricting to X_0 gives $E|_{X_0}$ trivial. By the isomorphism (4), $c_1(E) = 0$, hence E is trivial and $\mathfrak{s}'_1 \cong \mathfrak{s}'_2$.

For the equality of squares, represent the Poincaré dual of $c_1(L)$ by an oriented embedded surface $\Sigma \subset X$ (for instance, the zero locus of a transverse section of L). Since $\dim \Sigma + \dim \gamma = 3 < 4$, after a small perturbation we may assume $\Sigma \cap \gamma = \emptyset$, and after shrinking N we may assume $\Sigma \subset X_0$. Since $L'|_{X_0} \cong L|_{X_0}$, the same surface represents the dual of $c_1(L')$ in X' . Its self-intersection is computed entirely in X_0 , so it is unchanged by the surgery. This gives (3). $\square$

3.3 Change of expected dimension

The Euler characteristic changes by

$$\begin{aligned} \chi(X') - \chi(X) &= \chi(D^2 \times S^2) - \chi(S^1 \times D^3) \\ &= 2 - 0 = 2. \end{aligned}$$

The relative sequence for (X, X_0) also identifies $H^2(X; \mathbb{Z})$ with $H^2(X_0; \mathbb{Z})$: the map $H^3(X, X_0) \cong \mathbb{Z} \rightarrow H^3(X; \mathbb{Z})$ is the Poincaré dual of the primitive class $[\gamma]$ and is injective. Representatives of

degree-two classes can be chosen away from γ , so their pairings, and hence the real intersection form, are preserved. Thus

$$\sigma(X') = \sigma(X).$$

Combining this with Proposition 3.1 in the dimension formula (1), we obtain

$$d(X', \mathfrak{s}') = d(X, \mathfrak{s}) - 1.$$

Thus the hypothesis $d(X, \mathfrak{s}) = 1$ implies

$$d(X', \mathfrak{s}') = 0.$$

This already suggests that surgery should act as one cut-down condition; compare the dimension calculation in [1, Proposition 3.7].

4 Cylindrical ends and the gluing theorem

The analytic input will be used as a theorem, rather than reproved here. This avoids hiding the difficult points of gluing inside a short informal argument. I first define every object occurring in the statement and then quote the form of the theorem needed for the surgery formula. The local and global gluing results behind this statement are developed in Chapter 4 of Nicolaescu's notes, especially Theorems 4.5.7, 4.5.15, and 4.5.17 [5, Chapter 4]; the version allowing the reducible simple piece used below is Qiu's Theorem 3.13 [1, Theorem 3.13].

4.1 Cylindrical-end notation

Let M be a compact oriented Riemannian 4-manifold whose boundary is the closed oriented 3-manifold Y . Assume that the metric, the Spin^c structure, and the perturbation are products on a collar $[0, \varepsilon) \times Y$. The associated cylindrical-end manifold is

$$\widehat{M} = M \cup_Y ([0, \infty) \times Y).$$

The restriction of the Spin^c structure to Y is denoted by $\mathfrak{t}$. Write

$$\widehat{\mathcal{M}}(M) = \widehat{\mathcal{M}}(\widehat{M}, \mathfrak{s}_M)$$

for the moduli space of finite-energy solutions of the perturbed Seiberg–Witten equations on $\widehat{M}$, modulo gauge transformations which have a limit on the cylindrical end. Write

$$\mathcal{M}(Y) = \mathcal{M}(Y, \mathfrak{t})$$

for the moduli space of solutions of the three-dimensional Seiberg–Witten equations on Y .

For the positive-scalar-curvature boundary and the nondegenerate normal operators used below, every finite-energy monopole $\widehat{C}$ in the relevant moduli spaces converges exponentially, after a gauge transformation, to a limiting monopole C_∞ on Y . Its gauge-equivalence class depends only on $[\widehat{C}]$, so there is an asymptotic-value map

$$\partial_\infty : \widehat{\mathcal{M}}(M) \longrightarrow \mathcal{M}(Y), \quad [\widehat{C}] \longmapsto [C_\infty]. \quad (5)$$

This convergence and the construction of (5) are part of the cylindrical-end theory in [5, Sections 4.2–4.4].

At a finite-energy monopole $\widehat{C}$, the gauge-fixed linearization of the equations has a deformation complex. Its cohomology groups will be denoted by

$$H_{\widehat{C}}^0, \quad H_{\widehat{C}}^1, \quad H_{\widehat{C}}^2.$$

Here $H_{\widehat{C}}^0$ is the Lie algebra of the stabilizer, $H_{\widehat{C}}^1$ is the formal tangent space to the moduli space, and $H_{\widehat{C}}^2$ is the obstruction space. Let $F_{\widehat{C}}$ be the same deformation complex with the limiting boundary monopole fixed. The monopole $\widehat{C}$ is called *strongly regular* when

$$H^2(F_{\widehat{C}}) = 0. \quad (6)$$

The long exact sequence comparing the two deformation complexes then shows that $\widehat{C}$ is regular and that ∂_∞ is a submersion at $[\widehat{C}]$ [5, Section 4.3.2].

Now let M_1 and M_2 have the same boundary Y , with opposite boundary orientations, and compatible boundary Spin^c structures. For $r > 0$, form the closed manifold $M(r)$ by truncating the two ends and inserting the neck $[-r, r] \times Y$. A pair $([\widehat{C}_1], [\widehat{C}_2])$ is *compatible* if its two limiting monopoles agree. Thus the space of compatible gauge-equivalence classes is the fiber product

$$\widehat{\mathcal{M}}(M_1) \times_{\mathcal{M}(Y)} \widehat{\mathcal{M}}(M_2) = \{(x_1, x_2) : \partial_\infty x_1 = \partial_\infty x_2\}. \quad (7)$$

Cutoff functions glue a compatible pair to an approximate monopole on $M(r)$. After imposing the Coulomb slice condition, the nonlinear equation near this approximate solution has a finite-dimensional Kuranishi model

$$\kappa_r : B_\varepsilon(H_r^+) \longrightarrow H_r^-. \quad (8)$$

The space H_r^+ is the virtual tangent space and H_r^- is the obstruction space; they arise from the small-eigenvalue spectral subspaces of the self-adjoint extension of the glued, gauge-fixed linearization. The map κ_r measures the component of the nonlinear equation in the latter space after its orthogonal complement has been solved by a right inverse. If $H_r^- = 0$, then κ_r has zero target and the gluing is unobstructed. This is the content of the local gluing theorem [5, Theorem 4.5.7].

4.2 The gluing theorem used here

The following is the precise specialized result that will be used. It is quoted from Qiu's adaptation of the ordinary gluing theory [1, Theorem 3.13 and Corollary 3.15].

Theorem 4.1 (Specialized global gluing theorem). *Let $M(r)$ be obtained by gluing the cylindrical-end manifolds $\widehat{M}_1$ and $\widehat{M}_2$ along their common end Y . Assume the following.*

- (G1) *Y is S^3 or $S^1 \times S^2$ and its metric has positive scalar curvature.*
- (G2) *$b^+(M_1) > 0$ and $b^+(M_2) = 0$, where $b^+(M_i)$ is the dimension of a maximal positive subspace for the compactly supported intersection pairing on the interior of M_i .*
- (G3) *Every finite-energy monopole on $\widehat{M}_1$ is irreducible and strongly regular in the sense of (6).*
- (G4) *Every finite-energy monopole $\widehat{C}_2$ on $\widehat{M}_2$ is reducible, and $\dim_{\mathbb{R}} H_{\widehat{C}_2}^1$ is either 0 or 1.*
- (G5) *For every compatible pair, the glued obstruction space H_r^- in (8) is zero for all sufficiently large r .*

Then, for all sufficiently large r , every monopole on $M(r)$ is irreducible and regular, and the gluing locus obtained from compatible pairs is isotopic, inside the irreducible configuration quotient of $M(r)$, to the genuine moduli space $\mathcal{M}(M(r))$. When the boundary gauge transformations arising from M_1 and M_2 generate the full boundary gauge group, the quotient defining the gluing locus is the ordinary fiber product in (7), rather than a fiber product over a larger boundary quotient; hence

$$\mathcal{M}(M(r)) \cong \widehat{\mathcal{M}}(M_1) \times_{\mathcal{M}(Y)} \widehat{\mathcal{M}}(M_2).$$

The isotopy identifies the two moduli spaces as smooth manifolds. An equality of signed counts additionally requires the standard determinant-line orientation comparison and compatible homology orientations; that comparison is made in Section 6.3.

For the unperturbed positive-scalar-curvature boundary problem used here, the Weitzenböck identity makes the limiting critical set a circle of reducible flat connections, and finite-energy trajectories between them have zero curvature energy and are constant modulo gauge. This excludes the nonconstant neck trajectories (“tunneling”) that complicate general cylindrical-end compactness. Nicolaescu’s global theorem proves that the local gluing neighborhoods cover the entire long-neck moduli space [5, Theorems 4.5.15 and 4.5.17]. Qiu’s formulation isolates the remaining issue in condition (G5), which is necessary because one of the two monopoles in the surgery problem is reducible [1, Sections 3.3–3.4].

4.3 Verification for the surgery pieces

We now take

$$Y = S^1 \times S^2, \quad M_1 = X_0,$$

and take M_2 to be either

$$N = S^1 \times D^3 \quad \text{or} \quad N' = D^2 \times S^2.$$

All three boundary restrictions use the unique Spin^c structure on Y whose determinant line has first Chern class zero: $H^2(Y; \mathbb{Z}) \cong \mathbb{Z}$ has no 2-torsion, and every determinant restricted from N is trivial. The extension over N' has the same boundary restriction. Choose the product metric on Y and the standard positive-scalar-curvature metrics on the two simple pieces, cylindrical near the boundary. With perturbations zero on these simple pieces, the Weitzenböck formula and nonnegative scalar curvature force their finite-energy monopoles to have zero spinor; the remaining equations say that the limiting connections are flat. The metrics and end conditions used for this conclusion are part of the gluing setup in [1, Sections 2.3 and 3.3]. Their moduli spaces and boundary maps are

$$\mathcal{M}(Y) \cong \text{Hom}(H_1(Y; \mathbb{Z}), S^1) \cong S^1, \tag{9}$$

$$\widehat{\mathcal{M}}(N) \cong S^1, \quad \partial_\infty : \widehat{\mathcal{M}}(N) \xrightarrow{\cong} \mathcal{M}(Y), \tag{10}$$

$$\widehat{\mathcal{M}}(N') = \{p\}, \quad \partial_\infty : \widehat{\mathcal{M}}(N') \longrightarrow \mathcal{M}(Y), \quad p \longmapsto p. \tag{11}$$

In (9), the circle records the normalized holonomy of a flat Spin^c connection around $S^1 \times \{\text{pt}\}$. The map in (10) is the identity after using this holonomy coordinate, because the same loop generates $H_1(N; \mathbb{Z})$ and $H_1(Y; \mathbb{Z})$. The disk factor in N' kills that loop, so only one fixed boundary holonomy extends across N' , giving (11). Relative to the chosen reference connection this point can be called the trivial holonomy p . The point may rotate when that reference changes, but a regular fiber count is independent of this choice. These calculations are carried out in Qiu’s proof of the surgery formula [1, Sections 3.3 and 3.5].

Because the surgery loop represents the generator of $H_1(X; \mathbb{Z})$ modulo torsion, the restriction map

$$H^1(X_0; \mathbb{R}) \longrightarrow H^1(Y; \mathbb{R}) \quad (12)$$

is surjective. Since $b_2^+(X) > 1$, one can choose a compactly supported perturbation on $\widehat{X}_0$ for which every finite-energy monopole is irreducible and strongly regular. In particular,

$$\partial_\infty : \widehat{\mathcal{M}}(X_0) \longrightarrow \mathcal{M}(Y) \cong S^1 \quad (13)$$

is a submersion. The surjectivity in (12) allows a closed one-form on X_0 to adjust any boundary holonomy. Qiu first proves transversality for a fixed limiting connection and then translates the argument across the boundary circle; this produces one perturbation that works for all limits [1, Propositions 3.9 and 3.11]. The quoted analytic result, rather than the cohomological surjectivity alone, supplies strong regularity.

For N , the fixed-boundary obstruction vanishes. For N' , the fixed-boundary deformation complex has the one-dimensional topological term

$$\mathrm{im}(H^2(N'; \mathbb{R}) \longrightarrow H^2(Y; \mathbb{R})) \cong \mathbb{R}.$$

This fixed-boundary term is not by itself an obstruction to the closed problem: the boundary matching map sends its generator to the complementary boundary direction, with nonzero coefficient. Exactness of the linear gluing diagram therefore removes it. Establishing that map and controlling its small eigenvalues requires Qiu's calculation, not merely a count of dimensions. Consequently $H_r^- = 0$ in both gluings for all sufficiently large r [1, Propositions 3.11 and 3.12]. Thus conditions (G1)–(G5) of Theorem 4.1 hold for both decompositions.

5 The two global gluing identifications

For the remainder of the proof, choose a neck length for which Theorem 4.1 applies. We suppress the neck length from the notation: changing it gives a cobordant moduli space and does not change the Seiberg–Witten invariant.

5.1 Gluing back $S^1 \times D^3$

The theorem and (10) give

$$\begin{aligned} \mathcal{M}(X, \mathfrak{s}) &\cong \widehat{\mathcal{M}}(X_0, \mathfrak{s}_0) \times_{\mathcal{M}(S^1 \times S^2)} \widehat{\mathcal{M}}(S^1 \times D^3) \\ &\cong \widehat{\mathcal{M}}(X_0, \mathfrak{s}_0), \end{aligned} \quad (14)$$

because the boundary map from $\widehat{\mathcal{M}}(S^1 \times D^3)$ is the identity of the boundary circle. Under (14), define

$$\rho : \mathcal{M}(X, \mathfrak{s}) \longrightarrow S^1 \quad (15)$$

to be the map corresponding to (13). It is a submersion. Since the domain is a compact oriented 1-manifold, each of its connected components is a circle and the restriction of ρ to a component has a well-defined signed degree.

5.2 Gluing in $D^2 \times S^2$

Using (11), the same theorem gives

$$\begin{aligned} \mathcal{M}(X', \mathfrak{s}') &\cong \widehat{\mathcal{M}}(X_0, \mathfrak{s}_0) \times_{\mathcal{M}(S^1 \times S^2)} \widehat{\mathcal{M}}(D^2 \times S^2) \\ &\cong \partial_\infty^{-1}(p). \end{aligned} \tag{16}$$

Combining (14), (15), and (16) yields the precise geometric statement

$$\boxed{\mathcal{M}(X', \mathfrak{s}') \cong \rho^{-1}(p) \subset \mathcal{M}(X, \mathfrak{s})}. \tag{17}$$

This is exactly Corollary 3.15 and the first part of the proof of Theorem 3.17 in [1]. The displayed identification uses the existence, local uniqueness, and global coverage supplied by the quoted analytic gluing results. It is an identification for a suitable long-neck metric and perturbation; invariance then returns the result for the original closed manifolds.

6 Proof of the surgery formula

We now relate the circle-valued map ρ to the cohomology class Θ and compare the signed counts.

6.1 The boundary circle is the Θ direction

Recall that the class Θ is the degree-one class on the configuration quotient associated to the generator γ of $H_1(X; \mathbb{Z})$. It may be represented by the holonomy map

$$h_\gamma : \mathcal{B}(X, \mathfrak{s}) \rightarrow S^1.$$

On the gluing region, the limiting flat connection on $S^1 \times S^2$ is also completely determined by its holonomy around the S^1 factor. Because this is the same loop as γ in the common region, the boundary-holonomy map ρ and the restriction of h_γ to the moduli space have the same cohomology class. One sees this by moving the representative loop down the cylindrical collar: the resulting cylinder gives a homotopy of normalized holonomy maps [2, Proposition 9.1]. Their literal maps can differ with the reference connection and collar choices, so only the cohomological equality is needed. If

$$\iota : \mathcal{M}(X, \mathfrak{s}) \hookrightarrow \mathcal{B}^*(X, \mathfrak{s})$$

is the inclusion and η_{S^1} is the generator of $H^1(S^1; \mathbb{Z})$, then

$$\iota^* \Theta = \iota^* h_\gamma^* \eta_{S^1} = \rho^* \eta_{S^1}.$$

Therefore

$$\begin{aligned} \text{SW}_\Theta(X, \mathfrak{s}) &= \langle \iota^* \Theta, [\mathcal{M}(X, \mathfrak{s})] \rangle \\ &= \langle \rho^* \eta_{S^1}, [\mathcal{M}(X, \mathfrak{s})] \rangle \\ &= \deg \rho. \end{aligned}$$

Because $\mathcal{M}(X, \mathfrak{s})$ is a compact oriented 1-manifold, write

$$\mathcal{M}(X, \mathfrak{s}) = \bigsqcup_{i \in \Gamma} S_i^1.$$

Let

$$d_i = \deg(\rho|_{S_i^1} : S_i^1 \rightarrow S^1).$$

Then

$$\mathrm{SW}_\Theta(X, \mathfrak{s}) = \sum_{i \in \Gamma} d_i. \quad (18)$$

This is the precise meaning of cutting the moduli space down by Θ .

6.2 Counting the points after surgery

By (17), the moduli space on X' is the inverse image of the point p selected by the unique monopole on $D^2 \times S^2$. Since ρ is a submersion, p is a regular value, and on the component S_i^1 the inverse image consists of $|d_i|$ points counted with the sign of the local degree. Hence

$$\mathrm{SW}(X', \mathfrak{s}') = \#\mathcal{M}(X', \mathfrak{s}') = \sum_{i \in \Gamma} d_i. \quad (19)$$

Combining (18) and (19) gives the required equality

$$\mathrm{SW}_\Theta(X, \mathfrak{s}) = \mathrm{SW}(X', \mathfrak{s}').$$

6.3 Orientation and sign

The argument above identifies the underlying sets, but an equality of Seiberg–Witten invariants requires compatibility of orientations. The one-dimensional moduli space on X is oriented by a choice of homology orientation of

$$H^1(X; \mathbb{R}) \oplus H_+^2(X; \mathbb{R}).$$

The chosen generator of $H^1(X; \mathbb{Z})$ and a compatible homology orientation fix the sign of the circle class Θ . At a point $x \in \rho^{-1}(p)$ there is an exact sequence

$$0 \longrightarrow T_x \rho^{-1}(p) \longrightarrow T_x \mathcal{M}(X, \mathfrak{s}) \xrightarrow{d\rho} T_p S^1 \longrightarrow 0. \quad (20)$$

The zero-dimensional kernel is assigned the preimage orientation by requiring that the orientation of the middle term equal the ordered product of the kernel orientation and the positively oriented line $T_p S^1$. Thus a point receives the sign of the Jacobian of ρ in local oriented coordinates.

On the other hand, $\mathcal{M}(X', \mathfrak{s}')$ is oriented by the determinant line of the Seiberg–Witten deformation complex on X' . The determinant-line gluing isomorphism identifies this line with the determinant line on X after removing the one-dimensional boundary-holonomy direction. This is the cut-down orientation defined by (20), provided the homology orientation on X' is chosen by deleting the oriented H^1 summand generated by α from the one on X . Reversing this convention reverses one signed count. Therefore the identification

$$\mathcal{M}(X', \mathfrak{s}') \cong \rho^{-1}(p)$$

is orientation preserving, and the signed counts agree.

This completes the proof of Theorem 1.1.

6.4 Higher odd-dimensional moduli spaces

The same mechanism works when

$$d(X, \mathfrak{s}) = 2n + 1.$$

One defines

$$\mathrm{SW}_\Theta(X, \mathfrak{s}) = \langle \Theta \smile U^n, [\mathcal{M}(X, \mathfrak{s})] \rangle,$$

where $U \in H^2(\mathcal{B}; \mathbb{Z})$ is the usual degree-two class. The surgered manifold has

$$d(X', \mathfrak{s}') = 2n,$$

and its ordinary invariant is

$$\mathrm{SW}(X', \mathfrak{s}') = \langle U^n, [\mathcal{M}(X', \mathfrak{s}')] \rangle.$$

Provided the degree-two universal class U is defined using a base point in X_0 , the gluing identification respects its restriction to the common region. Its n -fold insertion cuts the positive-dimensional moduli spaces to a signed finite set; the boundary point condition supplies the remaining degree one. The same gluing and cut-down argument gives

$$\mathrm{SW}_\Theta(X, \mathfrak{s}) = \mathrm{SW}(X', \mathfrak{s}').$$

Thus the one-dimensional case is the cleanest instance of a general odd-to-even dimension drop. This higher-dimensional form is Qiu's Remark 3.18 [1].

7 Applications

The surgery formula is useful because ordinary Seiberg–Witten invariants on simply connected manifolds are already very effective. The formula transfers that information to manifolds with first homology $\mathbb{Z}$; the applications in this section follow Section 4 of [1].

7.1 Exotic smooth structures after adding $S^1 \times S^3$

Let Y be simply connected and set

$$X = Y \# (S^1 \times S^3).$$

Then

$$H^1(X; \mathbb{Z}) \cong \mathbb{Z}.$$

The generator is represented by the S^1 factor in the connected-sum summand. With the standard product framing, surgery on this generator changes $S^1 \times S^3$ to S^4 ; the connected sum with Y therefore recovers Y . The unique Spin^c structure $\mathfrak{s}_0$ on the summand has torsion determinant class. Since $\chi(Y \# (S^1 \times S^3)) = \chi(Y) - 2$ and the signature is unchanged, $d(X, \mathfrak{s} \# \mathfrak{s}_0) = d(Y, \mathfrak{s}) + 1$. In the zero-dimensional case $d(Y, \mathfrak{s}) = 0$, the surgery formula gives

$$\mathrm{SW}_\Theta(Y \# (S^1 \times S^3), \mathfrak{s} \# \mathfrak{s}_0) = \mathrm{SW}(Y, \mathfrak{s}). \quad (21)$$

In particular, ordinary zero-dimensional Seiberg–Witten invariants of Y become degree-one cut-down invariants of the nonsimply connected manifold $Y \# (S^1 \times S^3)$. If $d(Y, \mathfrak{s}) = 2n > 0$, the same assertion includes U^n on both sides as in Section 6.4. One must retain the Spin^c structure in this comparison: the symbol $\mathrm{SW}(Y)$ without it is generally a collection of coefficients, not a single integer.

The following transfer statement is Qiu's Theorem 4.1 [1].

Theorem 7.1. *Let Y_1 and Y_2 be simply connected closed oriented smooth 4-manifolds with $b_2^+(Y_i) > 1$. Suppose there is a Spin^c structure $\mathfrak{s}_1$ on Y_1 with a nonzero Seiberg–Witten invariant whose absolute value differs from $|\text{SW}(Y_2, \mathfrak{t})|$ for every Spin^c structure $\mathfrak{t}$ on Y_2 (using the appropriate U insertion when the expected dimension is positive). Then*

$$Y_1 \# (S^1 \times S^3) \quad \text{and} \quad Y_2 \# (S^1 \times S^3)$$

are not diffeomorphic.

Proof. Suppose there were an orientation-preserving diffeomorphism of the two connected sums. It would carry the chosen Spin^c structure to one on the second manifold, and $H^2(Y_i \# (S^1 \times S^3); \mathbb{Z}) \cong H^2(Y_i; \mathbb{Z})$ identifies that structure with one on Y_2 . Diffeomorphism invariance identifies their cut-down invariants, up to an overall sign coming from the homology orientation and the possible reversal of the generator of H^1 . Applying (21), or its U^n version, then gives equal absolute values for some invariant on Y_2 , contrary to the assumption. $\square$

This gives a direct way to transfer exotic smooth structures from simply connected to nonsimply connected 4-manifolds.

7.2 Elliptic surfaces and knot surgery

As recalled in Qiu’s proof of Corollary 4.2, Fintushel–Stern knot surgery supplies infinitely many exotic smooth structures in the situation needed here [1, Corollary 4.2]. Let $E(n)$ be an elliptic surface with $n > 1$ and let T be a regular torus fiber of self-intersection zero. Knot surgery along T produces manifolds $E(n)_K$ homeomorphic to $E(n)$ in this simply connected setting. Their Seiberg–Witten series is multiplied by the symmetrized Alexander polynomial $\Delta_K(t)$, with $t = \exp(2\text{PD}[T])$ under the usual series convention [3]. By choosing knots with different Alexander polynomials, one obtains infinitely many pairwise nondiffeomorphic smooth structures.

Now consider

$$E(n)_K \# (S^1 \times S^3).$$

Surgery on the generator of the $S^1 \times S^3$ factor recovers $E(n)_K$. For each Spin^c structure $\mathfrak{s}_K$ contributing a coefficient, the formula gives

$$\text{SW}_\Theta(E(n)_K \# (S^1 \times S^3), \mathfrak{s}_K \# \mathfrak{s}_0) = \text{SW}(E(n)_K, \mathfrak{s}_K).$$

Thus the entire collection of cut-down coefficients reproduces the Seiberg–Witten series of $E(n)_K$. Choosing knots whose Alexander polynomials give distinct coefficient data (for example, with unbounded maximal coefficient) distinguishes infinitely many smooth structures on the same underlying nonsimply connected topological manifold. This also explains why an equality written without a Spin^c argument should be understood coefficientwise.

Corollary 7.2. *For $n > 1$,*

$$E(n) \# (S^1 \times S^3)$$

admits infinitely many exotic smooth structures.

The point is not simply that one can form a connected sum with $S^1 \times S^3$. The new invariant is needed because the relevant moduli spaces on the resulting manifold are odd-dimensional. The surgery formula provides exactly the missing degree-one insertion.

7.3 Odd-dimensional basic classes

For an ordinary Seiberg–Witten invariant, a characteristic class

$$K = c_1(\det \mathfrak{s})$$

is called a basic class if the corresponding invariant is nonzero. In the present $b_1 = 1$ setting, Qiu extends this terminology to odd-dimensional moduli spaces [1, Definition 4.3].

Definition 7.3. Assume $H^1(X; \mathbb{Z}) \cong \mathbb{Z}$. A characteristic class $K = c_1(\det \mathfrak{s})$ is called a Seiberg–Witten basic class here if some Spin^c structure with this first Chern class satisfies one of the following conditions (this qualification matters when $H^2(X; \mathbb{Z})$ has 2-torsion):

- (1) for an integer $n \geq 0$, $d(X, \mathfrak{s}) = 2n$ and the pairing $\langle U^n, [\mathcal{M}(X, \mathfrak{s})] \rangle$ is nonzero; or
- (2) for an integer $n \geq 0$, $d(X, \mathfrak{s}) = 2n + 1$ and the pairing $\langle \Theta \smile U^n, [\mathcal{M}(X, \mathfrak{s})] \rangle$ is nonzero.

In the one-dimensional case this simply means

$$\text{SW}_\Theta(X, \mathfrak{s}) \neq 0.$$

The surgery formula turns such an odd-dimensional basic class into an ordinary basic class on X' . Following [1, Definition 4.4], call the $b_1 = 1$ manifold *of simple type* in this discussion when every nonzero cut-down invariant occurs in expected dimension one. By the higher-dimensional surgery formula, every nonzero ordinary invariant on X' then has expected dimension zero, so X' has ordinary Seiberg–Witten simple type. This implication explains the additional hypothesis in the negative-square clause below.

7.4 Adjunction inequality

Let $\Sigma \subset X$ be an embedded oriented connected homologically nontrivial surface. Assume first

$$[\Sigma]^2 \geq 0.$$

Let $K = c_1(\det \mathfrak{s})$ be an odd-dimensional basic class, say $d(X, \mathfrak{s}) = 2n + 1$ with

$$\langle \Theta \smile U^n, [\mathcal{M}(X, \mathfrak{s})] \rangle \neq 0.$$

Choose the surgery loop γ to represent the generator of $H_1(X; \mathbb{Z})$. Since

$$\dim \gamma + \dim \Sigma = 1 + 2 < 4,$$

we can arrange by general position that

$$\gamma \cap \Sigma = \emptyset.$$

Therefore Σ lies entirely in the common region X_0 and survives as the same embedded surface in X' .

Let $\mathfrak{s}'$ be the unique extension to X' and let

$$K' = c_1(\det \mathfrak{s}').$$

Because the Spin^c structures agree over X_0 ,

$$\langle K', [\Sigma] \rangle = \langle K, [\Sigma] \rangle.$$

The normal bundle of Σ is also unchanged, hence

$$[\Sigma]_{X'}^2 = [\Sigma]_X^2, \quad g_{X'}(\Sigma) = g_X(\Sigma).$$

By the surgery formula,

$$\text{SW}(X', \mathfrak{s}') = \langle \Theta \smile U^n, [\mathcal{M}(X, \mathfrak{s})] \rangle \neq 0.$$

Thus K' is an ordinary Seiberg–Witten basic class on X' . For $g(\Sigma) > 0$ and nonnegative self-intersection, apply the ordinary adjunction inequality to the nonzero invariant on X' [2, Theorem 1.4, with surface insertion 1]; it gives

$$2g(\Sigma) - 2 \geq [\Sigma]_{X'}^2 + |\langle K', [\Sigma] \rangle|.$$

Replacing the quantities on X' by the equal quantities on X yields

$$\boxed{2g(\Sigma) - 2 \geq [\Sigma]^2 + |\langle K, [\Sigma] \rangle|}.$$

The resulting statement is Qiu’s generalized adjunction formula [1, Theorem 4.5].

Theorem 7.4 (Adjunction inequality for odd-dimensional basic classes). *Let X be a closed oriented smooth 4-manifold with $b_2^+(X) > 1$ and $H^1(X; \mathbb{Z}) \cong \mathbb{Z}$, and let K be a Seiberg–Witten basic class in the extended sense above. If $\Sigma \subset X$ is an embedded oriented connected homologically nontrivial surface with $[\Sigma]^2 \geq 0$, then*

$$2g(\Sigma) - 2 \geq [\Sigma]^2 + |\langle K, [\Sigma] \rangle|.$$

If X is of simple type in the sense just defined and $g(\Sigma) > 0$, the same inequality holds without assuming $[\Sigma]^2 \geq 0$. Indeed, the surgery formula makes X' of ordinary simple type, and the negative-square adjunction theorem applies there [4]. This last assertion uses that external theorem; it does not follow from the nonnegative-square argument alone.

For genus zero and nonnegative square, the usual sphere-vanishing consequence of the adjunction theory rules out a nonzero basic class in the stated setting. For negative square the additional simple-type clause invokes a separate external result; it is not a consequence of disjointness from the surgery circle alone. The proof for positive genus and nonnegative square shows why the inequality has exactly the same form as the classical one. There is no new adjunction estimate on the surgered side. Instead, the surgery formula converts an odd-dimensional basic class on X into an ordinary basic class on X' , and the surface is untouched because the surgery is performed away from it.

8 The earlier surgery identity and its wider scope

The same surgery relation appears, with more general insertions, in Proposition 2.2 of Ozsváth and Szabó [2]. Their Proposition 2.2 assumes, in its own words, “a homologically trivial embedded two-sphere” [2, p. 7, Proposition 2.2]. Its operation removes $D^2 \times S^2$ and glues in $S^1 \times D^3$. It is therefore the inverse of the loop surgery considered above, once the new core sphere is shown to be homologically trivial. We make this qualification explicit before applying their result. In this section X need not have $b_1(X) = 1$.

8.1 Why a primitive loop gives a trivial core sphere

Let X be a closed oriented smooth 4-manifold, let $\gamma \subset X$ be an embedded framed loop whose class is primitive in $H_1(X; \mathbb{Z})/\text{tors}$, and set

$$X_0 = X \setminus \text{int } \nu\gamma, \quad X_\gamma = X_0 \cup_{S^1 \times S^2} (D^2 \times S^2).$$

Write $S = \{0\} \times S^2 \subset X_\gamma$. The restriction $H^1(X_0; \mathbb{Z}) \rightarrow H^1(S^1 \times S^2; \mathbb{Z})$ is surjective: a class pairing to 1 with $[\gamma]$ restricts to the generator on the boundary circle. The relevant part of the sequence of the pair (X_γ, X_0) is

$$H^1(X_0; \mathbb{Z}) \longrightarrow H^2(X_\gamma, X_0; \mathbb{Z}) \cong \mathbb{Z} \longrightarrow H^2(X_\gamma; \mathbb{Z}) \longrightarrow H^2(X_0; \mathbb{Z}) \longrightarrow H^3(X_\gamma, X_0; \mathbb{Z}) = 0.$$

Under excision and Poincaré–Lefschetz duality, the relative group is generated by the Thom class of the core sphere S . The first arrow evaluates a class of $H^1(X_0; \mathbb{Z})$ on the boundary circle and is surjective by primitivity. The image of the relative generator in $H^2(X_\gamma; \mathbb{Z})$ is $\text{PD}_{X_\gamma}[S]$. Thus $[S] = 0$ integrally, and restriction gives $H^2(X_\gamma; \mathbb{Z}) \cong H^2(X_0; \mathbb{Z})$. Comparing with the analogous sequence for $X = X_0 \cup (S^1 \times D^3)$ also identifies $H^2(X; \mathbb{Z})$ with $H^2(X_0; \mathbb{Z})$. Consequently a Spin^c structure on X has a unique corresponding extension across X_γ , as in Section 3.

If $[\gamma]$ is merely nonzero in rational homology but is divisible, the first arrow need not be onto. The core sphere may then represent nonzero torsion, and the uniqueness assertion and Proposition 2.2 cannot simply be applied in the form stated here.

8.2 The formula with insertions

Write $\mathcal{A}(X) = \mathbb{Z}[U] \otimes \Lambda^*(H_1(X; \mathbb{Z})/\text{tors})$, with $\deg U = 2$ and $\deg \mu(\beta) = 1$ for a loop class β . Here U is the point insertion and multiplication of degree-one classes uses the exterior sign rule. For a homogeneous element a of degree $d(X, \mathfrak{s})$, $\text{SW}_{X, \mathfrak{s}}(a)$ denotes the corresponding moduli-space pairing, extended linearly and taken to be zero in other degrees. The quotient $H_1(X; \mathbb{Z})/\text{tors} \rightarrow H_1(X_\gamma; \mathbb{Z})/\text{tors}$ induces a map $q : \mathcal{A}(X) \rightarrow \mathcal{A}(X_\gamma)$ taking U to U . We use the same letter for a class in H_1 and its degree-one insertion when no confusion arises.

Theorem 8.1 (Ozsváth–Szabó, in the loop-surgery direction). *Suppose $b_2^+(X) > 1$ and $[\gamma]$ is primitive as above. For every Spin^c structure $\mathfrak{s}$ on X , let $\mathfrak{s}_\gamma$ be the unique extension of $\mathfrak{s}|_{X_0}$ to X_γ . Then*

$$d(X_\gamma, \mathfrak{s}_\gamma) = d(X, \mathfrak{s}) - 1,$$

and, for every $a \in \mathcal{A}(X)$,

$$\boxed{\text{SW}_{X, \mathfrak{s}}(a \mu(\gamma)) = \text{SW}_{X_\gamma, \mathfrak{s}_\gamma}(q(a))}. \quad (22)$$

The two sides use compatible homology orientations; reversing one of these orientations changes the sign. As usual, an invariant on an insertion whose degree does not equal the expected dimension is understood to be zero.

Proof. The preceding pair-sequence argument gives $[S] = 0$, so Proposition 2.2 of [2] applies to the sphere surgery $X_\gamma \rightsquigarrow X$. In the notation of that proposition, the manifold containing the sphere is X_γ , the manifold containing its new core circle is X , and the core circle is γ . Its dimension statement and its identity

$$\text{SW}_{X, \mathfrak{s}}(a \mu(\gamma)) = \text{SW}_{X_\gamma, \mathfrak{s}_\gamma}(q(a))$$

are exactly the two claims. The map q spells out how their notation a is interpreted on the two manifolds. A lift of an insertion from X_γ to X is unique up to a multiple of $\mu(\gamma)$; multiplication by the

displayed extra $\mu(\gamma)$ makes the result independent of that choice. Their statement specifies a suitable homology orientation on the circle-surgery side; this accounts for the sign convention in (22). The theorem is invoked as an external surgery result: the pair-sequence proves the hypothesis on the core sphere, while its analytic gluing and orientation comparison are supplied by [2, Proposition 2.2]. $\square$

For $H^1(X; \mathbb{Z}) = \mathbb{Z}$, the chosen primitive loop determines $\Theta = \mu(\gamma)$. If $d(X, \mathfrak{s}) = 2n + 1$, substituting $a = U^n$ in (22) gives

$$\langle \Theta \smile U^n, [\mathcal{M}(X, \mathfrak{s})] \rangle = \langle U^n, [\mathcal{M}(X_\gamma, \mathfrak{s}_\gamma)] \rangle.$$

For $n = 0$ this is Theorem 1.1; for $n > 0$ it is the version discussed in Section 6.4 and in Qiu's Remark 3.18. Proposition 2.2 of [2] precedes [1] and allows $b_1(X) > 1$ and insertions besides powers of U . It does not by itself address the analytic details of Qiu's separate gluing proof or a surgery on a nonprimitive loop.

8.3 Several loops and the applications

Suppose disjoint framed loops $\gamma_1, \dots, \gamma_m$ represent part of an integral basis of $H_1(X; \mathbb{Z})/\text{tors.}$ Because their classes extend to part of an integral basis, quotienting by one retains the primitivity of the remaining classes. The sphere created at each stage is nullhomologous by the same pair-sequence argument. Thus the previous theorem applies successively. Iterating Theorem 8.1, with compatible choices of orientations, yields

$$\text{SW}_{X, \mathfrak{s}}(a \mu(\gamma_1) \cdots \mu(\gamma_m)) = \text{SW}_{X_{\gamma_1, \dots, \gamma_m}, \mathfrak{s}'}(q(a)),$$

where the expected dimension falls by m . For example, if $X = Y \# m(S^1 \times S^3)$, Y is a closed manifold with $b_2^+(Y) > 1$, the loops are the standard summand circles, and $d(Y, \mathfrak{s}_Y) = 2n$, then

$$\text{SW}_{Y \# m(S^1 \times S^3)}(U^n \mu(\gamma_1) \cdots \mu(\gamma_m)) = \text{SW}_Y(U^n),$$

with the structure on the left obtained by extending $\mathfrak{s}_Y$ over the standard summands and with compatible homology orientations. The dimension on the left is $2n + m$, exactly the degree of its insertion. This extends the transfer in Section 7.1 to several summands and is the mechanism used in the examples immediately following Proposition 2.2 of [2]. The nonvanishing of these cut-down invariants can distinguish smooth structures after adding several $S^1 \times S^3$ summands, subject to the same identification of Spin^c structures and homology orientations as above.

The adjunction discussion also has an earlier and, in some situations, sharper counterpart. Theorems 1.3 and 1.4 of [2] apply to a positive-genus surface Σ representing a non-torsion class with $[\Sigma]^2 \geq 0$. Their Theorem 1.4 gives the familiar bound

$$[\Sigma]^2 + |\langle c_1(\mathfrak{s}), [\Sigma] \rangle| \leq 2g(\Sigma) - 2$$

from any nonzero invariant $\text{SW}_{X, \mathfrak{s}}(a)$ when $b_2^+(X) > 1$: take the surface insertion $b = 1$ in that theorem. In particular, one can take $a = U^n \mu(\gamma)$, so this conclusion does not require first passing to the surgered manifold. Theorem 1.3 can improve the estimate if an insertion comes from the surface and its homological hypothesis holds.

For a concrete version, suppose $H^1(X; \mathbb{Z}) = \mathbb{Z}$, Σ has positive genus, and $\gamma = i_*(\beta)$ modulo torsion for some $\beta \in H_1(\Sigma; \mathbb{Z})$, where $i : \Sigma \hookrightarrow X$. Since the image of $H_1(\Sigma; \mathbb{R})$ in $H_1(X; \mathbb{R})$ is at most one-dimensional, its kernel contains a Lagrangian subspace. As the image contains the primitive class of γ , one can choose a symplectic integral basis whose A -cycles map to zero over

$\mathbb{R}$ and whose first B -cycle maps to γ modulo torsion. Thus $\ell = g(\Sigma)$ in [2, Theorem 1.3]; taking $a = U^n$ and $b = B_1$ gives a degree-one surface insertion with $1 \leq \ell$. The original class β may be used instead whenever it is itself a suitable first B -cycle. If

$$\mathrm{SW}_{X,\mathfrak{s}}(U^n\mu(\gamma)) \neq 0,$$

that theorem gives the stronger inequality

$$\boxed{[\Sigma]^2 + |\langle c_1(\mathfrak{s}), [\Sigma] \rangle| + 2 \leq 2g(\Sigma) - 2.}$$

The extra 2 requires a positive-genus surface and a degree-one insertion from its H_1 together with the stated symplectic-basis condition. The value of Θ alone supplies none of these hypotheses. For other surface insertions, the general statements and their different homological hypotheses are precisely [2, Theorems 1.3 and 1.4]. Their positive-genus, nonnegative-square assumptions should also be kept distinct from the negative-square simple-type clause in Qiu’s Theorem 4.5: Proposition 2.2 alone does not prove that clause.

In summary, for a primitive loop the $b_1 = 1$ formula proved by gluing in [1] is also a consequence of the earlier reverse sphere-surgery identity [2, Proposition 2.2], once the primitive-loop topology and the $\mu(\gamma) = \Theta$ normalization are made explicit. The stronger adjunction estimates come from additional theorems in [2], rather than from the surgery identity alone.

References

- [1] Haochen Qiu, *A Surgery Formula for Seiberg–Witten Invariants*, arXiv:2409.02265v1, 2024.
- [2] P. Ozsváth and Z. Szabó, *Higher Type Adjunction Inequalities in Seiberg–Witten Theory*, J. Differential Geom. **55** (2000), no. 3, 385–440; arXiv:math/0005268.
- [3] R. Fintushel and R. J. Stern, *Knots, links, and 4-manifolds*, Invent. Math. **134** (1998), 363–400.
- [4] P. Ozsváth and Z. Szabó, *The symplectic Thom conjecture*, Ann. of Math. (2) **151** (2000), 93–124.
- [5] L. I. Nicolaescu, *Notes on Seiberg–Witten Theory*, Graduate Studies in Mathematics, vol. 28, American Mathematical Society, 2000.